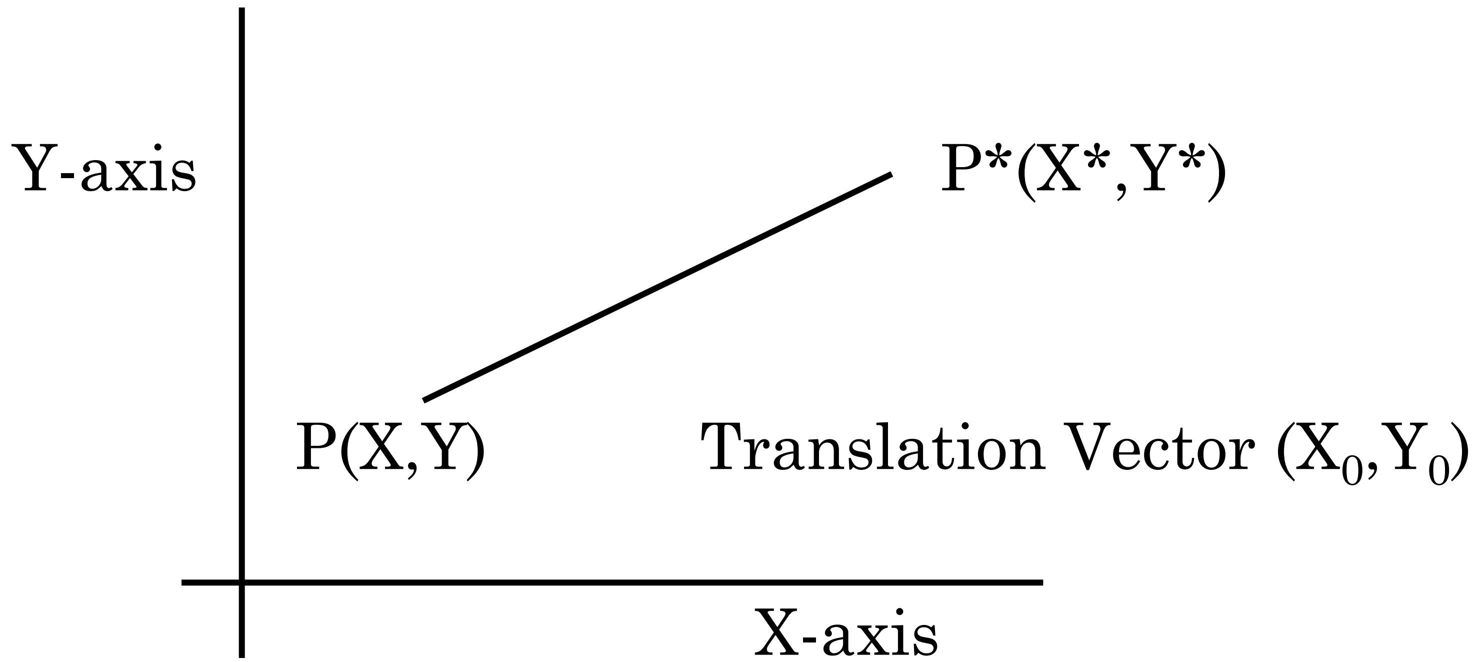
Image Geometry

By,
Prof (Dr) P P Ghadekar

OUTLINE

- On completion student will be able to understand
- Basic transformation-Translation, Rotation and Scaling in both 2-D and 3-D Image .
- Inverse transformation.

WHAT IS TRANSLATION OPERATION



Then, $X^* = X + X_0$

$Y^* = Y + Y_0$

Image Translation in 2-D

$$\begin{bmatrix} X^* \\ Y^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} + \begin{bmatrix} X_0 \\ Y_0 \end{bmatrix}$$

Above matrix equation can be represented in Single matrix form

$$\begin{bmatrix} X^* \\ Y^* \end{bmatrix} = \begin{bmatrix} 1 & 0 & X \\ 0 & 1 & Y \end{bmatrix} * \begin{bmatrix} X_0 \\ Y_0 \\ 1 \end{bmatrix}$$

IMAGE TRANSLATION IN 2-D

- Translation matrix in symmetric form

$$\begin{bmatrix} X^* \\ Y^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & X_0 \\ 0 & 1 & Y_0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

Above expression also called as Unified expression

IMAGE TRANSLATION IN 3-D

- Suppose that the task is to translate a point with coordinates (X,Y,Z) to new location by using displacements (X_0,Y_0,Z_0). The translation is easily accomplished by using the equations.

$$X^*=X+X_0$$

$$Y^*=Y+Y_0$$

$$Z^*=Z+Z_0$$

Where X^*,Y^*, Z^* are the coordinates of the new location. Above equation in matrix form is -

$$\begin{bmatrix} X^* \\ Y^* \\ Z^* \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & X_0 \\ 0 & 1 & 0 & Y_0 \\ 0 & 0 & 1 & Z_0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

-----(I)

Image translation

- Square matric representation of the matrix is

$$\begin{bmatrix} X^* \\ Y^* \\ Z^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & X_0 \\ 0 & 1 & 0 & Y_0 \\ 0 & 0 & 1 & Z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

--------(II)

In terms of the values of X^*,Y^*, Z^* are equivalent in equations (I) and (II).

IMAGE TRANSLATION

We can also used the unified matrix representation as given below

V*=TV

Where T is 4*4 transformation matrix, V is column vector containing the original coordinates and V* is the column vector whose components are transformed coordinates.

$$\mathbf{v} = \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \qquad \mathbf{v}^* = \begin{bmatrix} X^* \\ Y^* \\ Z^* \\ 1 \end{bmatrix} \qquad \mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & X_0 \\ 0 & 1 & 0 & Y_0 \\ 0 & 0 & 1 & Z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

So that above equation becomes **V*=TV**

Image scaling

- Scaling by factor S_X, S_Y and S_Z along the X, Y, and Z axes is given by transformation matrix

$$\mathbf{S} = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Image rotation

- To rotate a point about another arbitrary point in space requires three transformations as given below

I. Translates the arbitrary point to the origin.

II. Performs the rotation

III. Translates the point back to its original position.

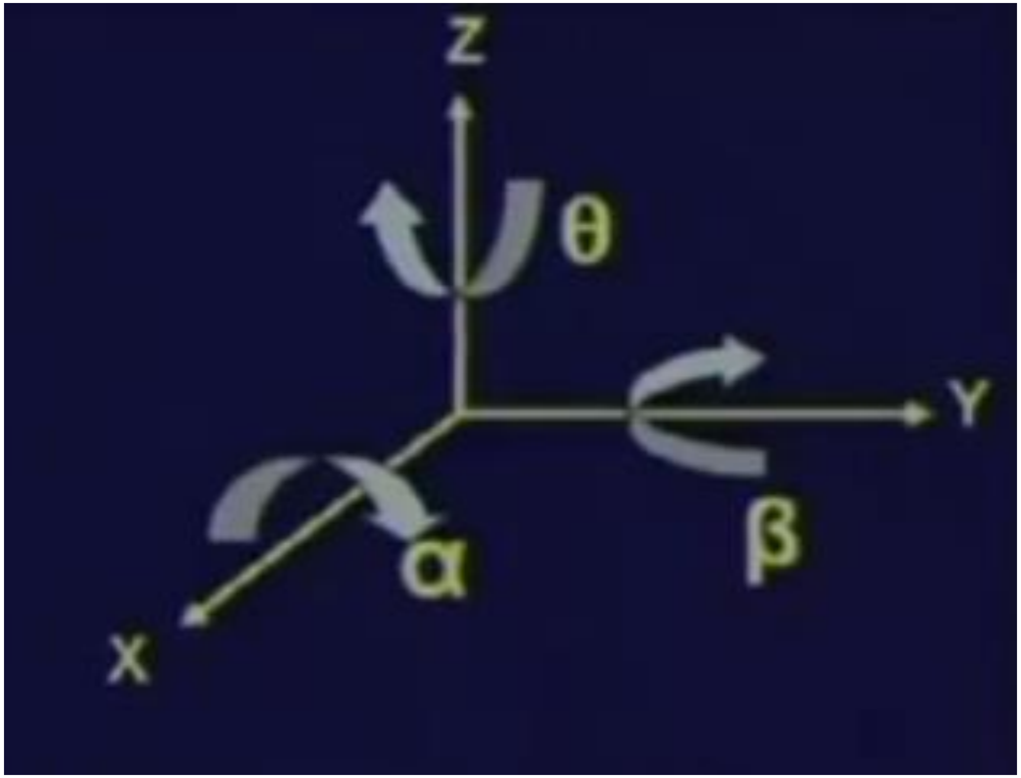


IMAGE ROTATION

- Rotation of a point about Z-axis by angle θ is achieved by using the transformation matrix given below.

$$\mathbf{R}_\theta = \begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- The rotation angle θ is measured clockwise when looking at the origin from a point on the Z-axis.
- This transformation affects values of X and Y coordinates.

Image rotation

- Rotation of a point about the X-axis by an angle α is performed by using the following transformation.

$$\mathbf{R}_\alpha = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha & 0 \\ 0 & -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

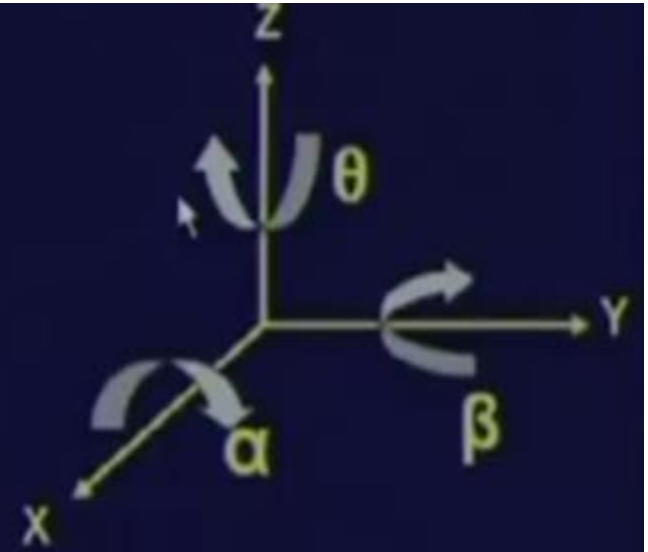
Image rotation

- Rotation of a point about the Y-axis by an angle β is performed by using the following transformation.

$$\mathbf{R}_\beta = \begin{bmatrix} \cos\beta & 0 & -\sin\beta & 0 \\ 0 & 1 & 0 & 0 \\ \sin\beta & 0 & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Image rotation

$$R_\theta = \begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$R_\beta = \begin{bmatrix} \cos\beta & 0 & -\sin\beta & 0 \\ 0 & 1 & 0 & 0 \\ \sin\beta & 0 & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_\alpha = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha & 0 \\ 0 & -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

INVERSE TRANSFORMATION

- Inverse translation matrix is given below.

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & 0 & 0 & -X_0 \\ 0 & 1 & 0 & -Y_0 \\ 0 & 0 & 1 & -Z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Similarly the inverse rotation matrix is given below

$$\mathbf{R}_\theta^{-1} = \begin{bmatrix} \cos(-\theta) & \sin(-\theta) & 0 & 0 \\ -\sin(-\theta) & \cos(-\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

INVERSE TRANSFORMATION

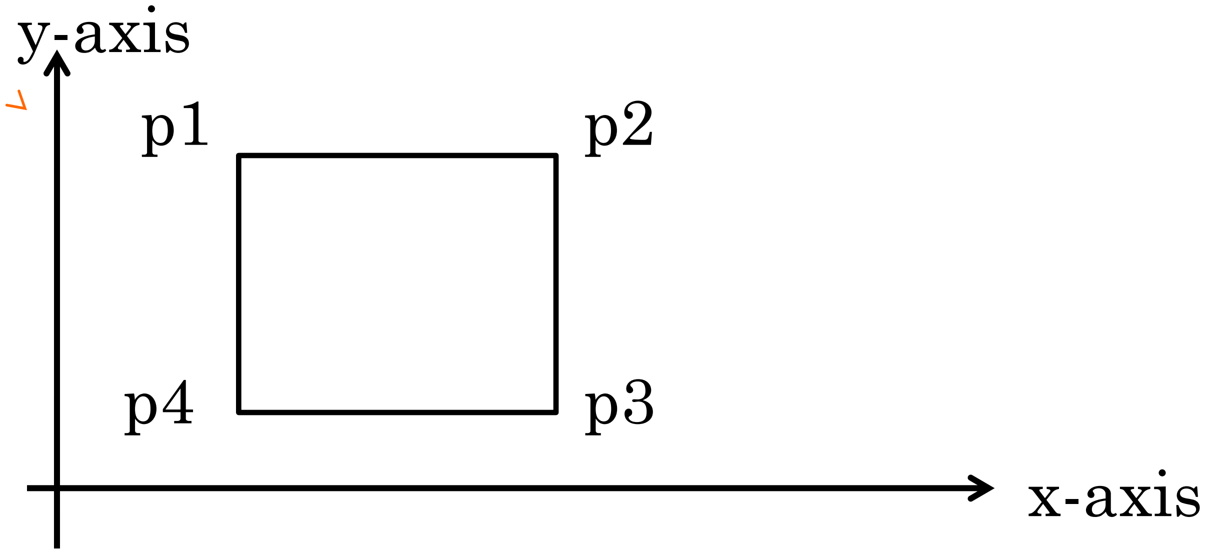
Can be Obtained by Observations

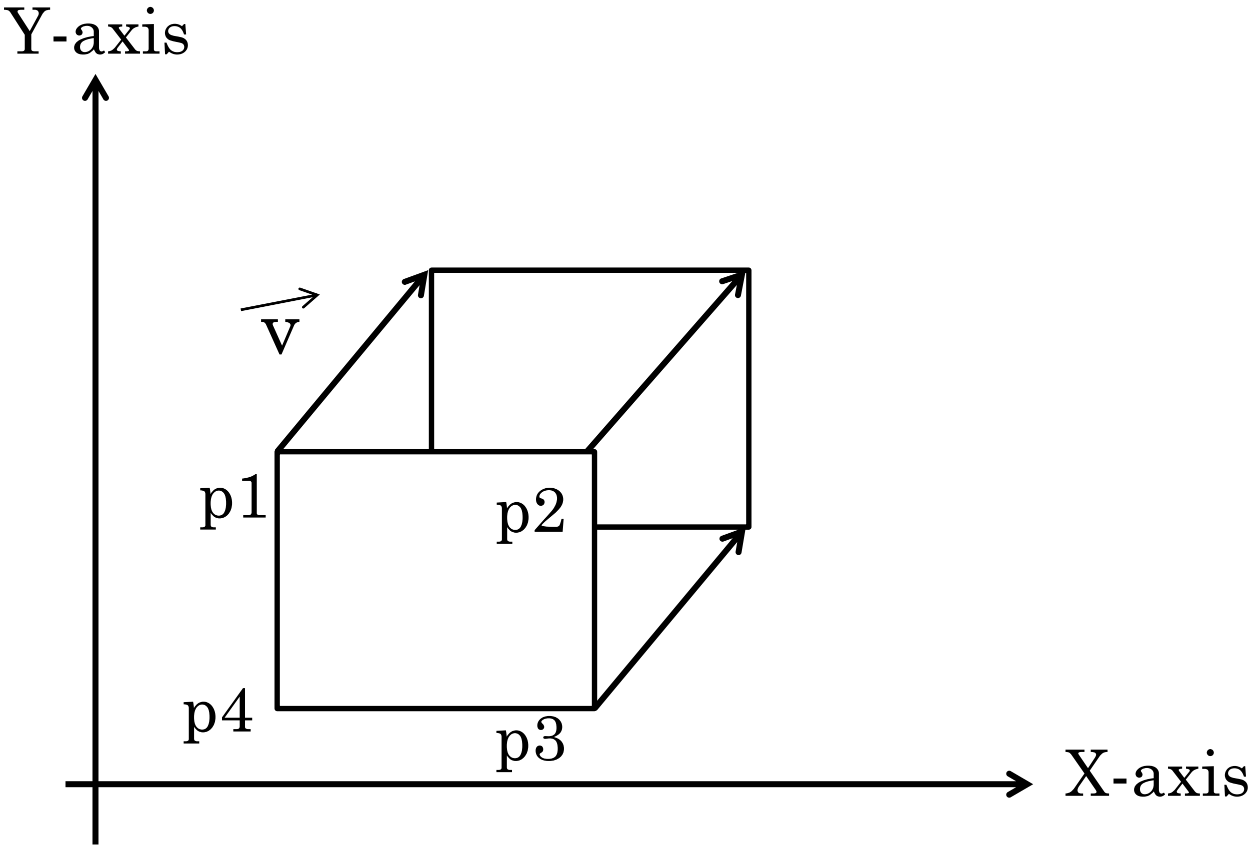
$$T^{-1} = \begin{bmatrix} 1 & 0 & 0 & -X_0 \\ 0 & 1 & 0 & -Y_0 \\ 0 & 0 & 1 & -Z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad R_\theta^{-1} = \begin{bmatrix} \cos(-\theta) & \sin(-\theta) & 0 & 0 \\ -\sin(-\theta) & \cos(-\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Transformation for a set of points

- For a set of m-points construct a matrix- V of dimension 4 x m.

The Transformation $V^* = TV$




Y-axis
v
p1
p2
p4
p3
X-axis

Thanks